

Development of User-Friendly Software Application for Extracting Information from National Bridge Inventory Source Files

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The nation is facing enormous repair, maintenance, and replacement costs for its aging bridge infrastructure, and this issue is of concern to a broad cross section of public and private interests. A simple-to-use bridge information system is needed by technical and nontechnical users who would like to review detailed information on the condition of the nation's public bridges without recourse to access-restricted and cost-prohibitive proprietary software. The annual National Bridge Inventory (NBI) text files collected from the states over consecutive years and maintained by FHWA constitute a wealth of information on approximately 600,000 public bridges throughout the United States, including information on condition and load ratings, geometry, sufficiency, age, location, functional classification, average daily traffic, improvement costs, inspection frequency, material and design types, historical significance, structural deficiency, and functional obsolescence. This information is of potential significance to a wide variety of end users, including bridge design and maintenance personnel, departments of transportation executive management, cultural and environmental specialists, economic forecasting and financial personnel, and media, commercial interest, and public information officers. As a supplement to a proprietary bridge management system, the New Mexico Department of Transportation developed an independent bridge information software application to process NBI source files and to produce commonly requested formatted reports on various categories of bridge information, including structural deficiency, age, improvement costs, historical significance, and several others. Because the NBI text files are processed into a common database format, custom reports may be easily generated by using simple query tools.

Each spring, state departments of transportation (DOTs) and other owner agencies are required to submit information on public bridges under their jurisdiction to the federal government. Information is encoded into a standard National Bridge Inventory (NBI) edit-extract file in conformance with formatting requirements as documented in Report No. FHWA-PD-96-001, *Recording and Coding Guide for the Structure Inventory and Appraisal of the Nation's Bridges (Coding Guide)* (1). The guide has been prepared for use by states, federal agencies, and other agencies in recording and coding the data elements that make up the NBI database. A thorough inventory per-

mits provision of reports to Congress on the number and condition of the nation's bridges, as well as providing data necessary for the FHWA and the Military Traffic Management Command to identify and classify the Strategic Highway Corridor Network and its connectors for defense purposes.

Many owner agencies employ the use of proprietary bridge management systems (BMSs), the most prevalent of which seek to maximize return on investment by employing techniques to predict deterioration rates of individual bridge elements and by suggesting maintenance strategies to best allocate limited funding for bridge improvement activities. While a comprehensive BMS serves as a useful tool for agency bridge inspection, design, and maintenance personnel, its use is often restricted to limited authorized personnel. Furthermore, use of these systems often requires a significant amount of technical support, training, and expertise. A recent synthesis by NCHRP reports that of the approximately 40 state DOTs that employ the most commonly used BMS, half of these use the program for inventory purposes only and do not use the advanced features, and that of those that do use advanced features, many use only a small subset of these features (2).

Information submitted annually by owner agencies includes about 100 individual items including location, description, load ratings, material and design type, recommended maintenance and estimated improvement cost, year built, condition ratings, geometry, inspection date and frequency, and more. These coded items are an integral part of the database used to satisfy several federal reporting requirements, as well as part of the states' needs. These requirements are set forth in 23 CFR 650.3, the National Bridge Inspection Standards (3).

AASHTO's *Manual for Condition Evaluation of Bridges* discusses various items that are required to be recorded as part of original bridge reports (4). That manual and the *Bridge Inspector's Manual* (5) discuss inspection procedures and preparation of detailed reports that form the basis for recording values for many of the elements documented in the *Coding Guide*, particularly with those related to condition and appraisal ratings.

USES OF NBI INFORMATION

Owner agencies are the primary users of NBI information. Bridge inspection personnel from these agencies are required to collect and report the data elements documented in the *Coding Guide*, and bridge management staff use this information to identify and prioritize bridge maintenance, repair, and replacement activities. However, the NBI database constitutes a wealth of information of potential interest to a broad cross section of public and private users. These

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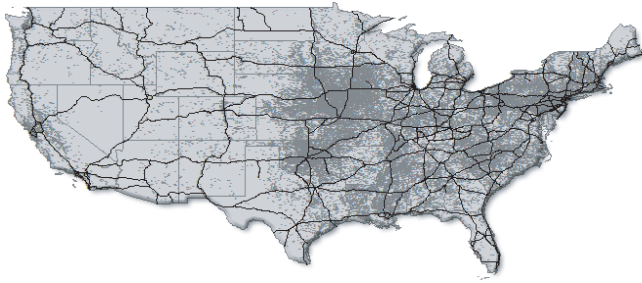


FIGURE 1 Distribution of structurally deficient bridges in contiguous United States. (Source: 2009 NBI database, FHWA.)

include, for example, university students and principal investigators, DOT executive staff, trucking industry personnel, cultural and environmental specialists, DOT project development engineers, economic forecasting and financial personnel, transportation researchers, media interests, private companies, and public information officers. In addition to identifying and prioritizing maintenance and repair needs, the information may be used for a variety of purposes including determining the location and condition of deficient bridges (Figure 1), performing trend studies using NBI data over consecutive years (Figure 2), evaluating compliance with inspection frequency requirements, estimating improvement cost escalation, locating load- and clearance-restricted bridges, and providing data necessary for evaluation of stresses imposed on bridges by overweight vehicles. Structural deficiency and functional obsolescence are directly coded in the NBI database, derived using NBI information in accordance with the definition of these terms as documented in FHWA *Non-Regulatory Supplement OPI: HNG-33* (6).

Because bridge maintenance, repair, and replacement needs far outstrip available funding, many states have adopted a BMS that seeks to maximize available funding by examining historical data on costs and deterioration rates of individual bridge elements as documented in AASHTO's *Commonly Recognized Elements (CoRE) Guide* (7), by employing statistical analysis to predict deterioration and preservation costs, and by suggesting maintenance, repair, and replacement scenarios given available budget. Limitations of commercial BMSs are annual licensing fee requirements, access restricted to a limited number of authorized users, information provided for only one state, and the requirement for a significant degree of technical support and expertise. By contrast,

NBI data are provided as public records and are readily accessible to the general public. These data, however, are provided in the form of encoded alphanumeric text files from which meaningful information is difficult to extract by nontechnical users. What is needed is an easy-to-use means by which this information may be accessed by users with minimal knowledge of industry practice and data coding conventions.

SABIS BRIDGE INFORMATION SYSTEM

While commercially available BMSs serve the needs of agency bridge inspection and management staff, their function is primarily directed toward managing bridge inventories, maintenance and repair activities, and budgets. The state-of-the-art bridge management system used by most state DOTs employs sophisticated statistical methods for predicting element-level deterioration, tracking improvement and preservation costs, and developing scenarios that permit agency personnel to maximize return on investment of limited funding. A simplified bridge information system based on NBI data elements, while lacking the sophistication and predictive capabilities of a commercial BMS, would nevertheless provide a low-cost supplemental application that serves many of the needs of other end users. Furthermore, because all states submit information in a common file format, and because this information is available for each year going back to 1992, such an application would be easily scalable to permit concurrent examination of various classifications of bridges from all states over consecutive years. This would allow user agencies and others to conduct network-level analyses and to perform comparative studies on various categories of bridges between states. Figure 3, for example, presents a comparison of the location of scour-critical bridges in four southwestern states. (Scour is a term used to describe erosion at structure supports due to the effects of moving water. Scour may have potentially catastrophic consequences, and a bridge is rated in the NBI database as scour-critical when the foundation is determined to be unstable for calculated scour conditions; when a field review has indicated that extensive scour has occurred; when a field review indicates that failure of a pier or abutment is imminent; or when a bridge has failed and is closed to traffic. Nationwide, approximately 20,000 bridges are classified as scour critical.)

To address the need for a user-friendly software application for extracting and processing information from NBI source files, the New Mexico Department of Transportation developed SABIS (Special Application—Bridge Information System). The program

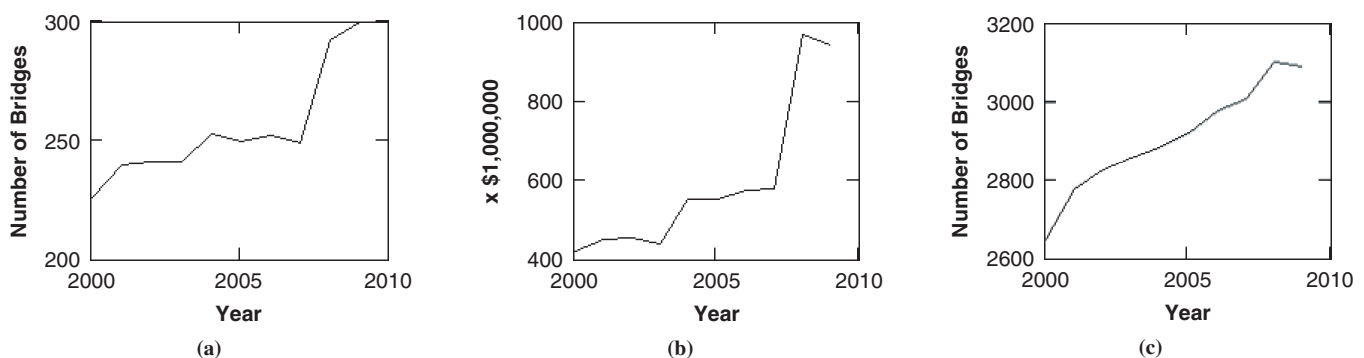


FIGURE 2 Trend studies for various categories of bridges in New Mexico bridge inventory: (a) sufficiency rating < 50, (b) bridge improvement cost, and (c) age > 25 years. (Source: 2000–2009 NBI databases, FHWA.)

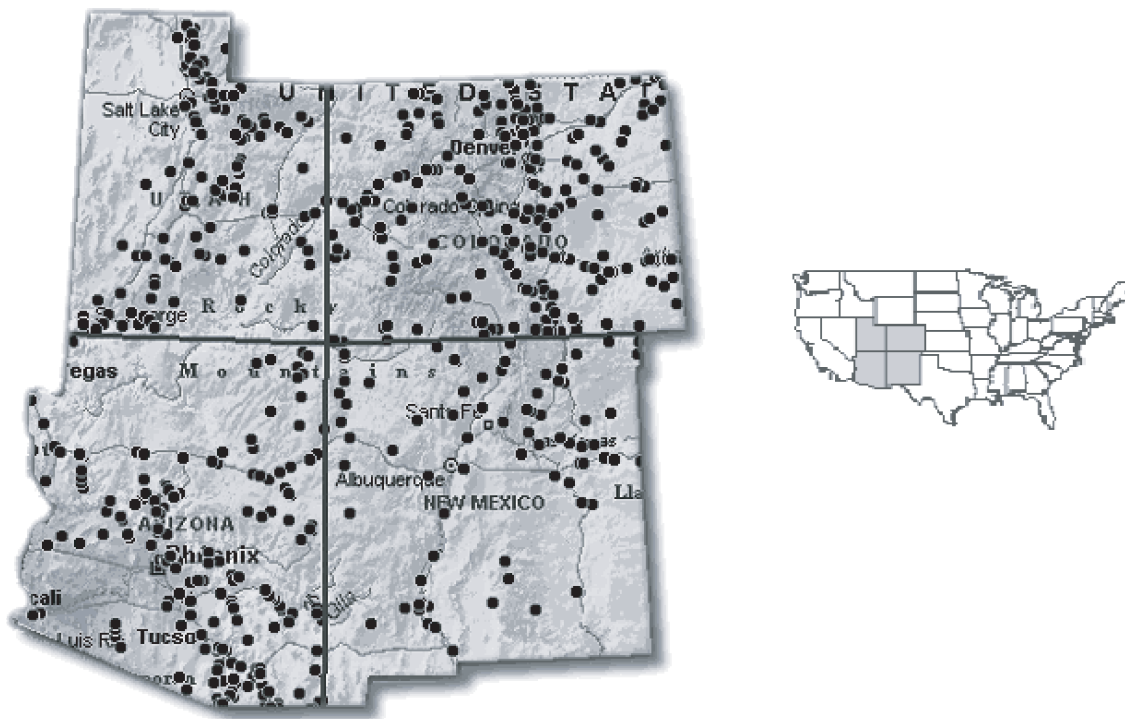


FIGURE 3 Comparison of location of scour-critical bridges in four southwestern states (Arizona, Colorado, New Mexico, Utah). (Source: 2009 NBI database, FHWA.)

is designed as a low-cost supplement to its BMS to be used by technical and nontechnical end users. A key concept in program development is that end users have varying degrees of knowledge and skills that may not include an understanding of industry practice and conventions used in the *NBI Coding Guide*. Accordingly, the program was developed to perform as much data processing and analysis as possible in the background and to provide an intuitive user interface by which users may approach the information with little or no specific knowledge of the bridge inventory or industry coding conventions (Figure 4).

Unlike commercial BMS programs, which have evolved over many years as comprehensive BMSs for use by professional practitioners in bridge inspection and inventory management, SABIS is strictly a tool for extracting and reporting information encoded into NBI edit-extract files. Because the software and source files are independent from an agency's BMS, use of SABIS has no effect on the management operations of the owner agency. The limitation of this is that the information extracted and processed by SABIS represents a "snapshot" of the bridge inventory at the time of the NBI file creation. The information is not real-time, and if SABIS users rely on publicly available NBI files as the data source, the information provided may be more than a year out of date. SABIS may therefore not be used as a means for making decisions or reaching conclusions related to bridge management, and owner agencies should be consulted if current information on a particular structure is needed. This limitation is of minor consequence to many end users who may not necessarily need up-to-the-minute information on bridges, as much of this information, including location, material and design type, roadway classification, historical significance, year built, geometry, and so forth, remains constant over successive inspection cycles. Further, because most of these structures are inspected on a 2-year

cycle, the bulk of the information encoded into the most recent NBI data files will be reasonably current.

Locating a Specific Structure

Each bridge in the inventory for each state is assigned a unique 15-digit alphanumeric identifier. While this identifier is meaningful to professional bridge inspection and management personnel, to most casual users it is not. Therefore, SABIS provides the means to locate a structure of interest in multiple ways based on general information. This includes identification of structure location by

- Bridge number,
- County,
- Material and design type,
- Route and milepost,
- Feature intersected, and
- Map click.

Although it is unlikely that a casual user will know a specific bridge number, he or she will generally have some limited information to narrow the search. For example, if the user knows that a bridge is a steel deck truss, the program may be queried for this material and design type, and a list of all steel deck trusses will be presented on a map and a pick list. The user may scroll through this list and review summary information on the bridge. This feature is useful, for example, in examining condition information on structures in a state's bridge inventory with design and material similar to the I-35 bridge in Minnesota that collapsed in August 2007. Similarly, if the only information known is that the bridge crosses the Rio Grande

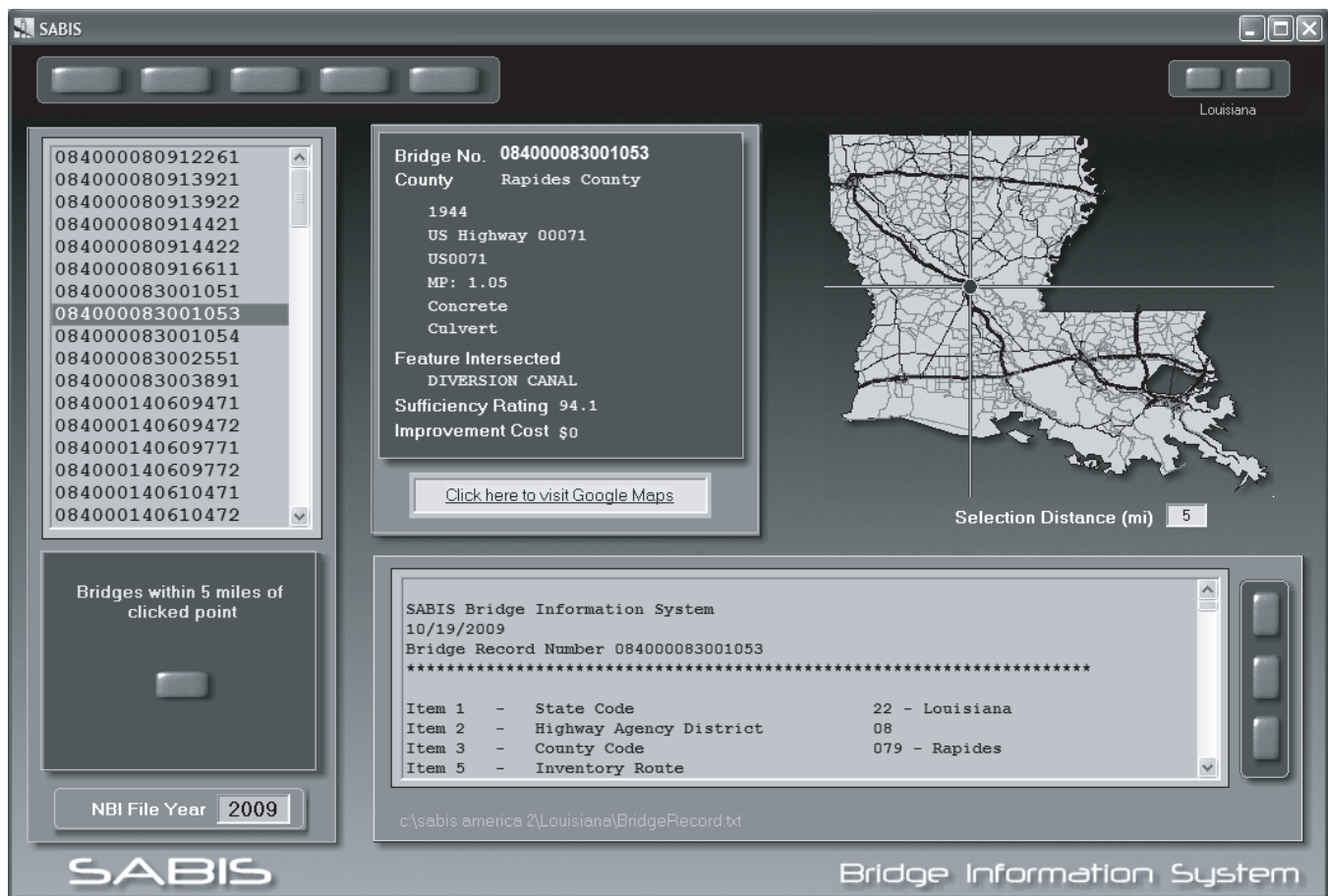


FIGURE 4 SABIS user interface, customized for use with Louisiana NBI source files.

River in New Mexico, that the bridge is somewhere in Montgomery County, Tennessee, or that the bridge is between mileposts 125 and 150 on Interstate 44 in Missouri, the program may be queried for this criteria and the user may select the structure of interest from a filtered list. If no information is known other than general location, the user may simply input a selection distance and click on the map. The program will display all structures in the inventory within a specified distance from the point selected on the map.

When a structure has been selected by the user, SABIS displays summary information and plots its location on a state map. The program also displays a detailed report that presents each of the NBI data items and, when appropriate, decodes the data item and performs unit conversion (Figure 4). In some cases, the user may desire to view the location of the bridge on detailed roadway, terrain, or satellite maps. SABIS provides a hyperlink to Google Maps that jumps directly to the point of interest (Figure 5). Google Maps displays the location on satellite, terrain, or roadway maps. In some instances, as in this example in Figure 5, a panoramic view of the location is provided and detailed examination of a particular structure may be performed.

In addition to bridge-specific reports, SABIS analyzes the entire NBI data file, plots all of the structures on a state map, and summarizes network-level statistics. This report presents information of general interest including, for example, the number of bridges on file, structurally deficient and functionally obsolete bridges sorted by roadway

functional classification, improvement cost, sufficiency ratings, and so forth (Figure 6).

Deficient Bridge Reports

Deficient bridges generate a considerable amount of interest, especially following a major structural event such as a bridge collapse. The term can be confusing, however, as state agencies and news organizations sometimes have different ways of defining the term. To minimize confusion, SABIS determines structural deficiency and functional obsolescence in accordance with the definitions for these terms as presented in FHWA's *Non-Regulatory Supplement OPI: HNG-33* (6). The NBI database directly codes bridges in these categories as derived from other NBI-coded items.

In addition to structural deficiency and functional obsolescence, other categories of bridges are of interest. The NBI database assigns a numerical appraisal rating for various categories of bridge information, including structural evaluation, deck geometry, horizontal and vertical underclearance, waterway adequacy, and approach roadway alignment. SABIS produces formatted reports for bridges rated as "intolerable," when a high priority of corrective action or replacement is warranted in these categories. Reports on scour-critical bridges, sufficiency ratings, bridges with undermined channel protection, improvement costs, and bridges overdue for inspection are

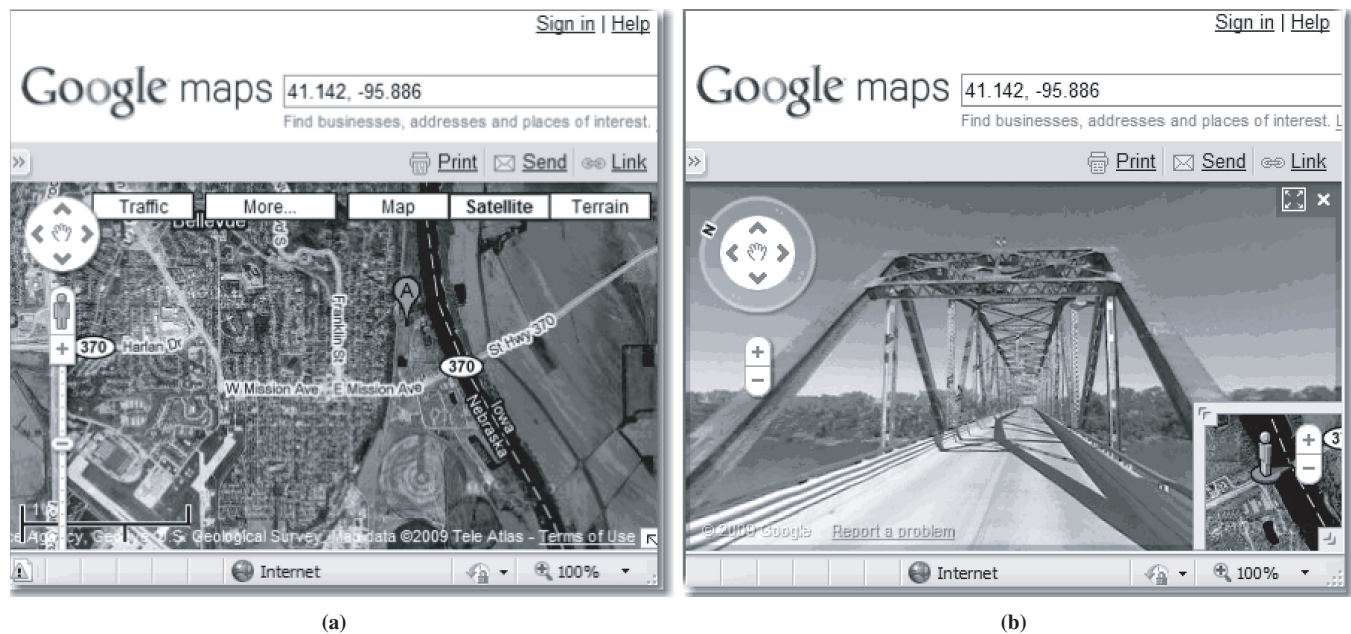


FIGURE 5 Bridge No. 36100 in Iowa: (a) satellite view and (b) street view. This is a low-sufficiency rating steel through-truss bridge on State Highway 370 over the Missouri River with repair cost of \$15 million. (Source: 2007 NBI database, FHWA. Map and photo courtesy of Google Maps.)

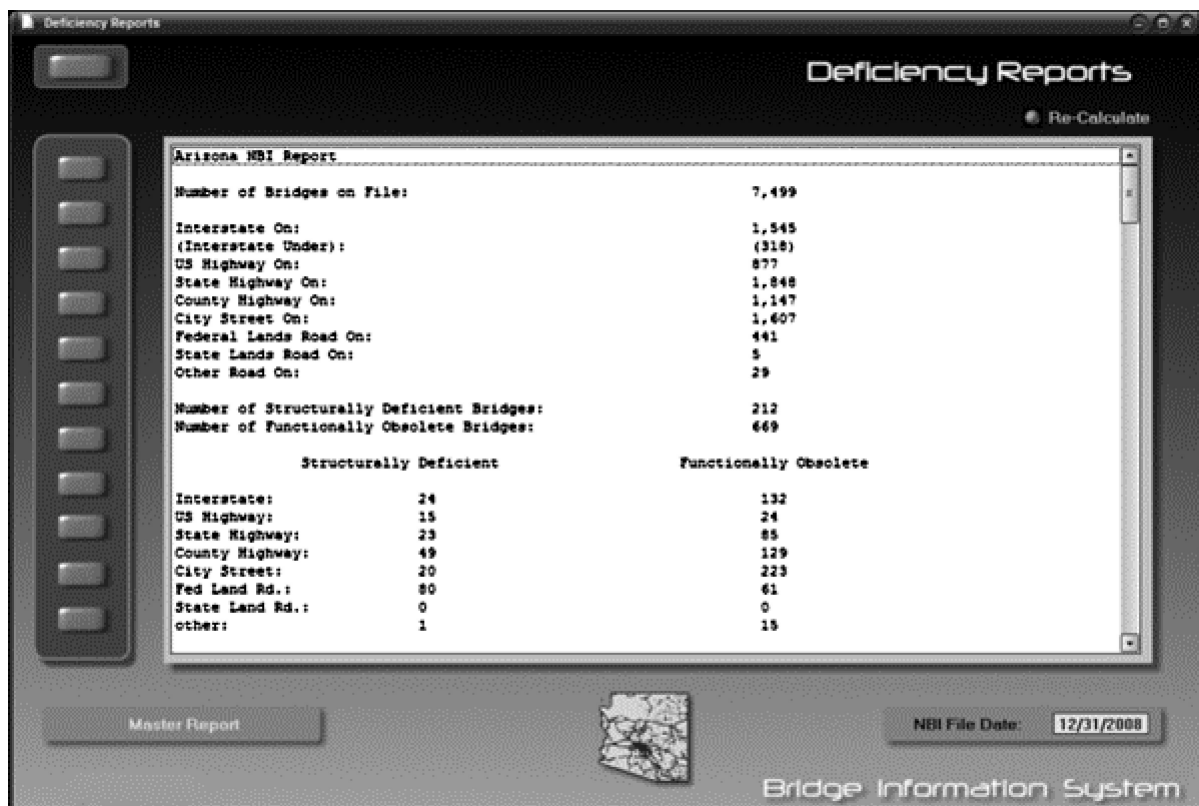


FIGURE 6 Arizona NBI summary report as extracted by SABIS. (Source: 2009 NBI database.)

also produced. These reports include bridge number, county, route and milepost, sufficiency rating, improvement cost, and latitude and longitude coordinates, and the reports are formatted so as to permit simplified plotting of structure locations using a variety of commonly available commercial and consumer grade mapping programs (Figure 7).

Multimedia

Multimedia files present a potentially significant means for conveying information. As a bridge information system, SABIS provides the means to organize and store multimedia files including digital bridge photos, Quicktime Virtual Reality files, and instructional videos, as well as the native capability to view these files. When a bridge is selected by the user, SABIS scans the multimedia directory for image files related to the bridge. If image files are detected, SABIS displays an option that permits the user to view these files (Figure 8).

Hyperlink Library

The Internet provides a vast amount of information, including countless websites related to bridges. Available information includes photos, instructional videos, statistics, raw data, scholarly dissertations, presentations, and much more. Some of these websites may be of interest to a particular group of users, while others may not.

As a means to conveniently manage these websites, SABIS provides a feature that permits users to add, delete, and access links to useful websites on Internet-enabled workstations. A current list of URLs and descriptive information is updated to a text file to allow SABIS users to share information on Internet searches and relevant websites with other interested users. This feature is similar to a Favorites folder provided by many web browsers (Figure 9).

Custom Queries

SABIS produces standard reports for information of general interest. These include detailed bridge-specific reports, summary statistics reports, deficient bridge reports, and reports on bridges overdue for inspection, improvement costs, and sufficiency ratings.

On occasion, users may want to produce custom reports based on information available in the NBI database. SABIS decodes the NBI text files and populates a Microsoft Access database with this information. Users with knowledge of *Coding Guide* conventions and a general understanding of filtering information in an Access database may use Access query tools to develop custom reports and maps. Figure 10 presents maps of Interstate bridges in Champaign County, Illinois, with intolerable underclearances, and Interstate bridges in Dallas County, Texas, with an inventory load rating less than HS17. (The NBI database codes two load ratings for each bridge. Inventory load rating is set at 55% of yield strength and represents the maximum load level that the bridge may be subjected to on a regular basis. Operating rating is set at 75% of yield strength and represents the ultimate load level to which the bridge may be subjected.) The following example illustrates this process:

Needed: Interstate bridges in Dallas County, Texas, with inventory load rating $\leq$ HS17.

Solution: NBI filter criteria are

1. Bridge in Dallas County (NBI Item 3 = "113"),
2. Bridge "on" an Interstate route (NBI Items 5A and 5B = "1"), and
3. Inventory load rating $\leq$ HS17 (NBI Item 66 $\leq [10 \cdot (17 \cdot 1.8 \cdot 0.9)] = "275"$).

When the database is filtered for these criteria, fields from the filtered list, including latitude and longitude, may be selected, dropped into a commercial spreadsheet, and saved as a tab-delimited text file. The resulting text file may then be used to plot locations of filtered

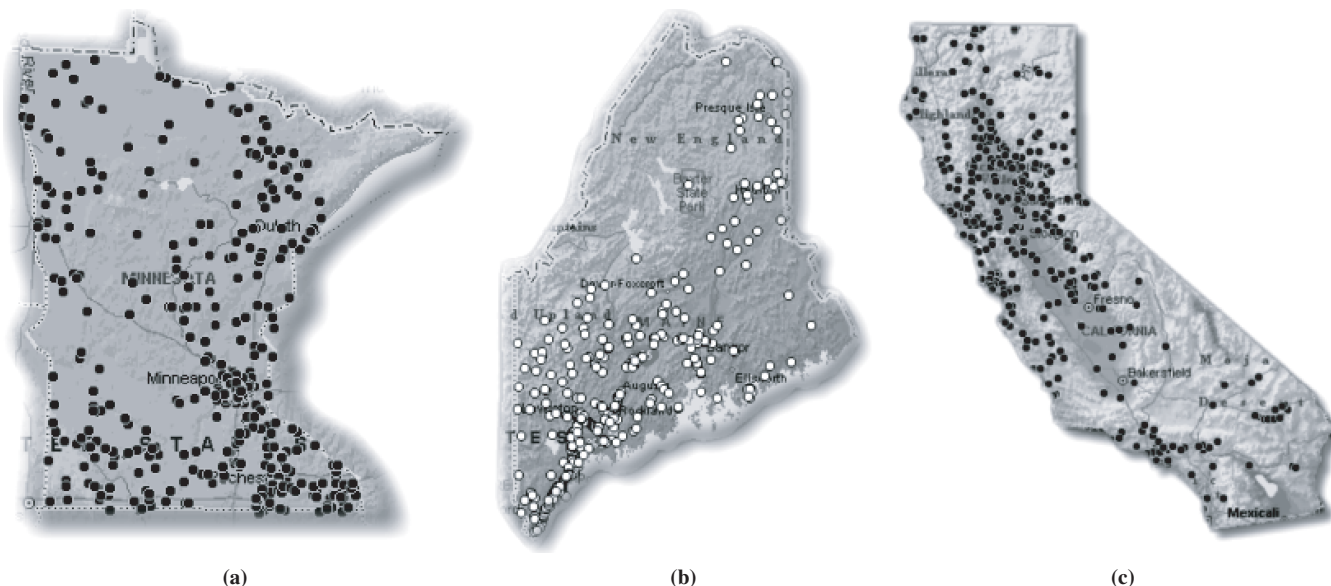


FIGURE 7 Random sample of plotted bridges in various categories: (a) bridges with intolerable deck geometry in Minnesota, (b) bridges with undermined channel protection in Maine, and (c) bridges with intolerable structural evaluation in California. (Source: 2009 NBI database, FHWA.)



FIGURE 8 SABIS provides the means to organize, store, and natively view multimedia files. Shown here is a historically significant steel lift bridge in Yolo County, California. (Photo courtesy of Google Maps.)

bridges on a variety of commercial and consumer grade mapping applications.

A variety of useful reports may be generated using simple database query tools, and more advanced analysis may be performed through computer programming. These reports range from review of overdue inspection data by agency management to evaluation of

bridges with poor deck ratings for consideration by manufacturers of deck overlay products. Users are cautioned, however, to be thoroughly familiar with NBI coding conventions to avoid erroneous or misleading results. Professional agency bridge management staff should be consulted when information presented in the NBI database is in doubt, and to ensure that the information is accurate and current.



FIGURE 9 A hyperlink library provided by SABIS allows users to organize, store, and share hyperlinks to bridge-related websites.

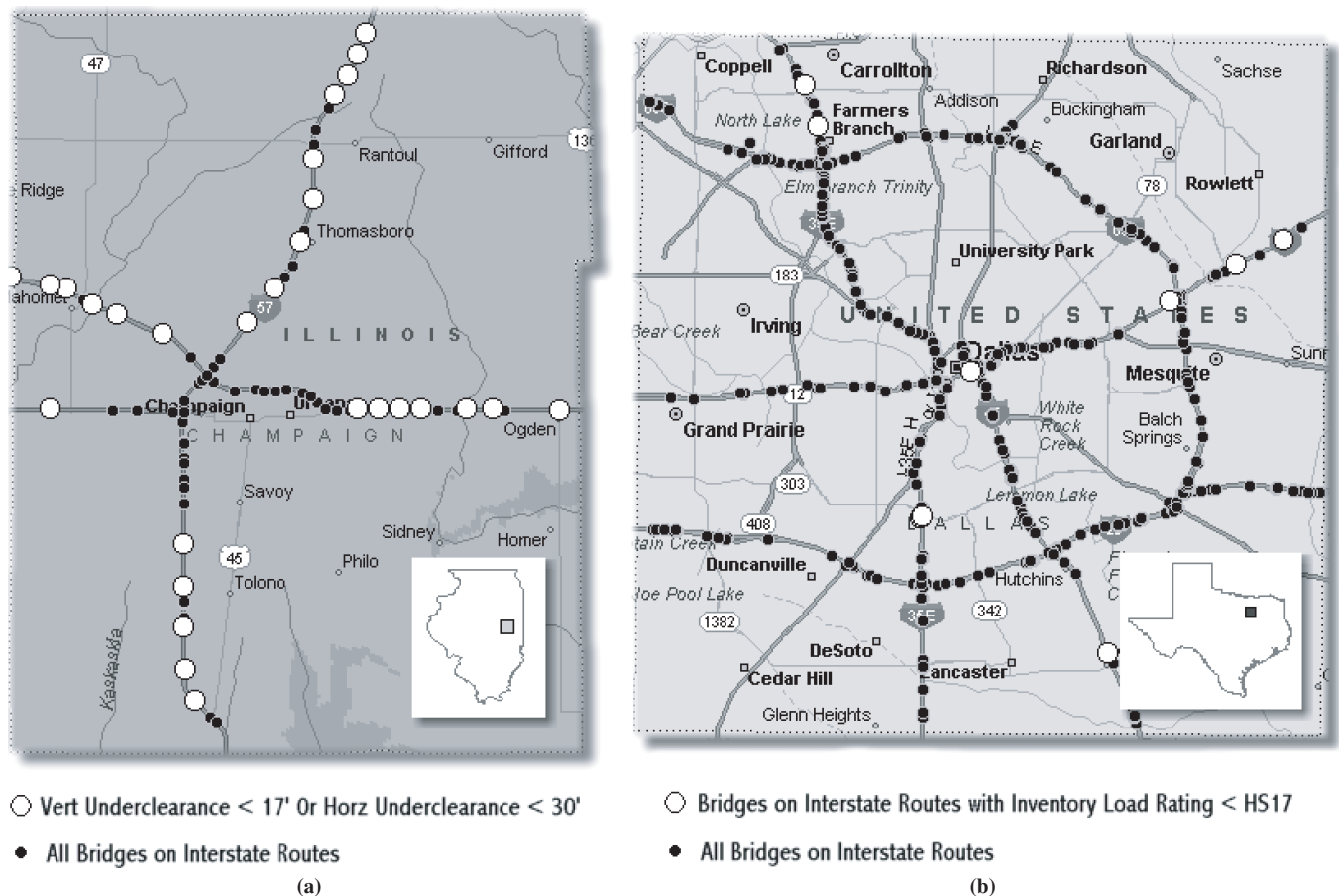


FIGURE 10 Interstate bridges: (a) in Champaign County, Illinois, with intolerable underclearances and (b) in Dallas County, Texas, with inventory load rating less than HS17. (Source: 2009 NBI database, FHWA.)

Because NBI source data do not reflect up-to-the-minute condition information, reports generated through SABIS should not be used for the basis of management decisions that require current data.

Use of NBI Database in Other Applications

In addition to serving as an independent source of useful information, data fields in the NBI database may be parsed for use in other applications. This may include evaluation of improvement cost escalation or growth of average daily traffic (ADT) over consecutive years, or analysis of the effect of overweight vehicles on the nation's bridges.

In the following example, information on an overweight vehicle as recorded by a state's weigh-in-motion equipment is used in conjunction with data from the NBI database and a simplified structural analysis routine to estimate allowable shear and moment stresses on a nearby bridge.

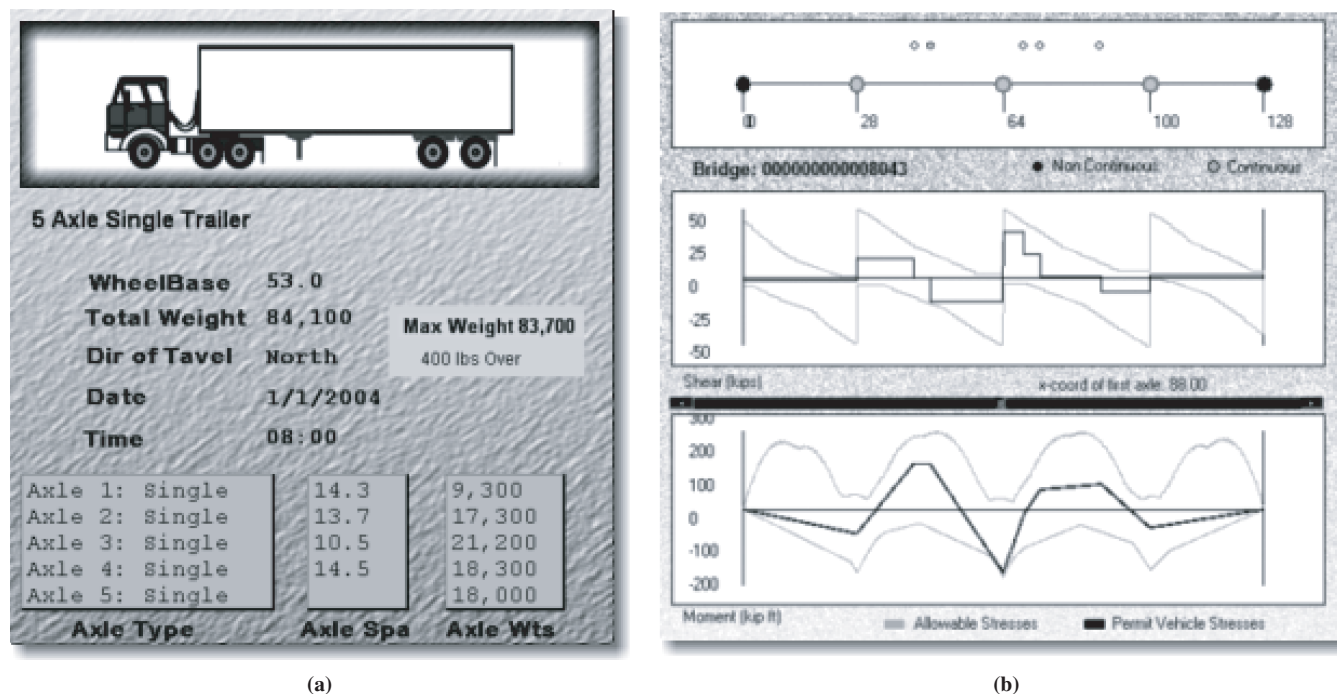
Highway bridges do not function in isolation, but rather they are an integral and vital component of an increasingly complex transportation system. Numerous factors affect bridge performance, condition, and longevity, including composition and volume of traffic, environmental pollution, condition of vehicles and driver behavior, to name a few. Figure 11 presents an example of how information from various data sources, including the NBI database and weigh-in-motion equipment, may be used conjointly to gain insight into various aspects of the transportation system and to permit

decision makers to make informed choices regarding the prudent use of diminishing resources.

SUMMARY

The aging and deteriorating bridge infrastructure presents significant challenges for current and future bridge management personnel. As shown in Figure 12, the mean and median ages of bridges less than 100 years old in the 2009 NBI database are 39 years. Erection of new bridges has steadily declined over the past decade, and within the next decade half of the bridges in the inventory will be over the commonly cited bridge service life threshold of 50 years. Current and future managers of the nation's bridge infrastructure face significant challenges in maintaining the population of aging structures with escalating repair costs and declining resources. A simple-to-use bridge information system may be useful in conveying information on the condition of the nation's bridge inventory to interested stakeholders from diverse backgrounds, whose technical, administrative, financial, and research expertise will be vital in effective future bridge management.

The NBI database represents a historical body of information that may be subject to statistical analyses by state DOTs, independent researchers, and other stakeholders. Analysis may be performed, for example, on deck, substructure, and superstructure performance with and without improvement actions, deck survival based on



wearing surface type, ADT, and time-in-condition state, among others. A simple-to-use application for processing NBI text files would streamline this analysis, as well as provide a rich source of information for those who may not be familiar with industry practice and coding conditions. SABIS provides this functionality.

Data Quality

To make effective and reliable use of NBI information, data quality issues must be addressed by owner agencies and other stakeholders. Data quality issues include gaps in the data record, inaccurate data entry, out-of-range values, and input codes not in conformance with

NBI coding conventions. Data quality issues are significant when drawing conclusions based on statistical analysis of NBI data, as some information, fluctuations in structure ratings for example, may be attributable to coding errors rather than assumed improvement activities. A significant example of coding error is evident in evaluating improvement costs. The NBI coding convention requires that the bridge improvement cost, a six-digit field in the database, be coded in thousands of dollars. On review of the NBI data, however, it is clear that there are numerous entries in which this cost is coded in actual dollars, resulting in a thousandfold erroneous increase in the reported cost. In one representative state, for example, a cost field in its 2009 NBI submittal is recorded as 774150, or \$774,150,000. A relatively small number of miscoded cost entries have the potential to

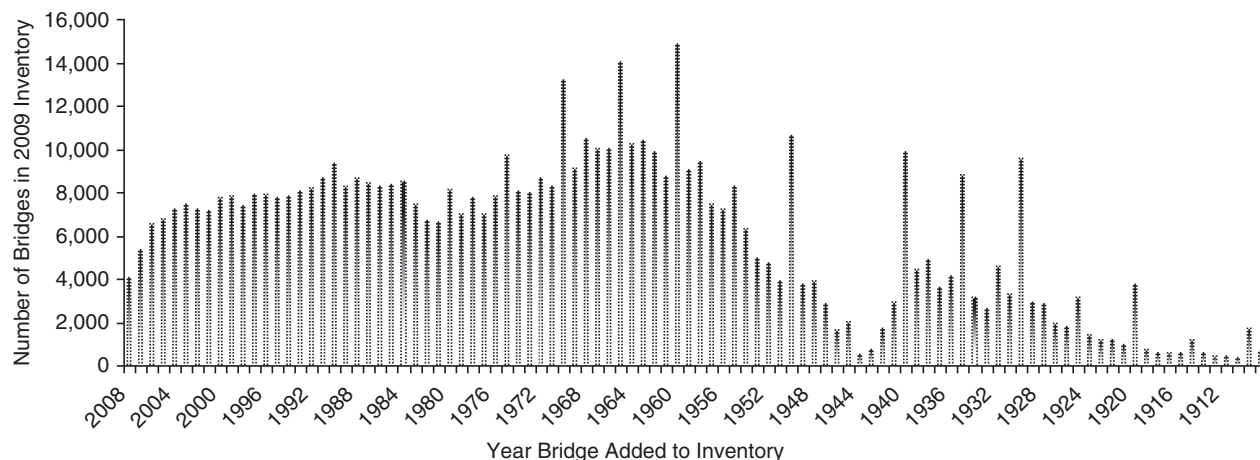


FIGURE 12 Age distribution of U.S. bridges less than 100 years old. The mean and median ages of bridges in this population are 39 years. (Source: NBI database, FHWA.)

significantly skew the overall reported cost of improvement. Similarly, approximately 5% of the inventory as a whole is lacking latitude and longitude data fields completely, and a significant percentage of bridges that are coded for these fields are coded incorrectly.

An important benefit of an application such as SABIS is that it provides an opportunity for a researcher to perform a “reasonableness check” on the data. A standard SABIS report, for example, provides a list of bridges sorted by improvement cost in descending order (Figure 13). Reviewing this list will allow a researcher to quickly identify entries that fail a subjective reasonableness check.

Figure 13 presents a portion of a report on a state’s bridge improvement cost in descending order. According to the 2009 NBI database, 0.1% of this state’s bridges account for two-thirds of the total recorded improvement cost. A skeptical researcher would suspect that these reported costs are grossly overestimated, and this may lead to diminution of confidence in the integrity of other data items as well. As a primary source of information, the issue of data quality in the NBI database is of significant concern, especially when these data are cited as a source in drawing conclusions and making decisions regarding the condition of the nation’s vital bridge infrastructure.

National Security Considerations

The NBI constitutes a major investment in the economy of the United States, and specific structures may present potential targets for foreign and domestic terrorists. Following the September 11, 2001, terrorist attacks, the FHWA removed NBI source files from its website and redacted some of the publicly available information normally provided through the NBI database. The policy at that time was to provide all data elements with the exception of four: latitude, longitude, critical facility indicator, and Strategic Highway Network indicator.

In May 2007 the FHWA released a memo referenced as HIBT-30 (8). That document outlines FHWA’s policy regarding the availability of NBI data. According to the memo, the NBI database does not present information not otherwise publicly available that would likely assist individuals in targeting a specific structure; withholding this information from public disclosure does not improve transportation security, nor does it provide additional protection for particular structures. The document further reaffirms the U.S. Department of Transportation’s policy that protection of potentially sensitive information must be balanced against the public’s interest in how governmental agencies carry out their responsibilities. The memo states that information concerning the condition of public bridges that also forms the basis for decisions regarding the expenditure of public funds is a matter of legitimate public interest, and that nondisclosure of NBI data does not improve transportation security. For these rea-

sons, the FHWA has concluded that its 2001 policy of withholding certain NBI data elements cannot be supported, and its current policy is one of full disclosure of all information contained in the NBI database.

CONCLUSIONS

As an informational tool, the NBI database is made up of detailed information on public bridges throughout the United States. The database supports FHWA bridge programs, including the Highway Bridge Program and the National Bridge Inspection Program. The data assist FHWA in meeting federal responsibilities and reporting requirements and assist the states in meeting national bridge inspection programs. Information in the database is collected and maintained with public funds.

In addition to assisting government agencies in satisfying statutory requirements and managing bridge inventories, the NBI database is a rich source of information to numerous public and private entities. In recognition of this, the FHWA provides this information to interested parties on its website. The information, however, is provided in the form of encoded alphanumeric text files, which are of limited value to users who may lack the means to process these files or who may be unfamiliar with industry practice and coding conventions.

SABIS provides the means to process publicly available NBI data files and to present this information to end users with limited knowledge of industry practice. Perhaps more importantly, the application parses data from NBI source files, performs unit conversions and other preprocessing tasks, and populates a simple single-table database that provides the information in a readily accessible format for any desired state. This allows researchers and other stakeholders to perform more advanced statistical analyses and to derive more customized information than may be provided in standard SABIS reports.

SABIS is intended as a low-cost supplement to a comprehensive BMS to permit end users with little or no technical knowledge to approach the significant body of information on public bridges as collected by government agencies over consecutive years and compiled in annual NBI data source files. Advantages of using a system like SABIS include

- Minimal user cost,
- No recurring licensing fees,
- Minimal technical knowledge required by end users,
- Ready availability of data over consecutive years,
- Use that is not restricted to a small group of authorized users,
- Information that is not restricted to a single state, and
- A simple database architecture that uses a commercially available application.

Br No	Cost	SR	Route	MP	County
000079301322476	\$774,150,000	72.0	County Highway 00849	007.1	Randolph
000082304018053	\$604,875,000	78.5	County Highway 00048	005.3	St. Clair
000060322509007	\$527,400,000	99.7	County Highway 00737	003.3	Madison
000082402318169	\$486,720,000	77.0	Other Road 09254	000.2	St. Clair
000014300003302	\$480,720,000	84.0	County Highway 00782	001.2	Clinton
000082306109429	\$463,200,000	56.6	Other Road 09148	001.7	St. Clair
000007000401707	\$414,285,000	22.2	State Highway 00100	045.2	Calhoun
000082026605604	\$368,790,000	80.0	State Highway 00159	031.0	St. Clair

FIGURE 13 Representative output from a standard SABIS report presenting bridge improvement costs in descending order. (Source: 2009 NBI database, FHWA.)



FIGURE 14 Subset of a state NBI database ported to a handheld device.

The open-source nature of the software permits interested parties to customize the application to meet particular needs. For example, the application could be customized by a state DOT to include a bridge inspection module ported to a handheld device with data uploaded to a central server. Figure 14 presents an example of a subset of NBI data used in a mobile application.

The disadvantage of a bridge information system that relies on publicly available data files is that the lag time between NBI file creation and publication means that the information will not be current. This is of minor consequence to casual users and researchers performing historical analysis of data over consecutive years; however, users are cautioned that the data are not real-time, and should not be relied on for making decisions and drawing conclusions when current information is required. Unlike the sophisticated BMS applications used by state agencies, SABIS does not provide information on element-level bridge details, nor does it perform management functions. SABIS is strictly a tool for reviewing the most recent information as provided through publicly available data files.

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The General Structures Committee peer-reviewed this paper.